

Autodesk® Scaleform®

擴展 **ActionScript** 參考

Scaleform 4.2

ActionScript 2.0

Artem Bolgar

4.03

2012 9 17

Copyright Notice

Autodesk® Scaleform® 4.2

© 2012 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AUGI, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo) Built with ObjectARX (design/logo), Burn, Buzzsaw, CAiCE, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, DesignServer, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWFx, DXF, Ecotect, Evolver, Exposure, Extending the Design Team, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, Freewheel, GDX Driver, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, RiverCAD, Robot, Scaleform, Scaleform GfX, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, Tinkerbox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

ActionScript 2.0

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

1	介紹.....	1
2	擴展滑鼠類.....	2
2.1		2
2.2		2
2.3		4
2.4		5
2.5	ActionScript	7
3	擴展按鈕類.....	10
4	MovieClip 擴展類	12
5	擴展 NetStream 類	16
6	擴展選擇類.....	19
7	擴展 TextField 類.....	27
7.1		27
7.2		35
7.3		41
7.4	HTML	43
7.5		45
7.6		49
7.7	IME	53
8	擴展 TextFormat 類.....	56
9	擴展 Stage 類	57
10	陣列類擴展.....	59
11	String 類擴展.....	60
12	視頻類擴展.....	61
13	3Di Extensions.....	62
14	全局擴展	63
15	標準方法和事件處理常式擴展.....	66

1

Autodesk Scaleform SDK
Adobe® Creative Suite®
Flash Web
PC
Scaleform
Adobe® Flash
IME
Scaleform
ActionScript
Scaleform Player
_global.gfxExtensions
true
_global.gfxExtensions = true;
FLA
Flash
Scaleform Player
Flash® Player
Scaleform
SDK
Scaleform
Scaleform Player
Scaleform
API
API

2

	Scaleform		
RIGHT, MIDDLE		16	
Scaleform			Flash Mouse Class

2.1

	Wii™	Scaleform 3.2	4
Gfx::Movie::SetMouseCursorCount(unsigned n)			
Gfx::Movie::HandleEvent with Gfx::MouseEvent		Scaleform	
Gfx::MouseEvent		Gfx::MouseEvent	

2.2

onRollOver, onRollOut, onDragOver, onDragOut, onPress, onRelease,			
onReleaseOutside	Scaleform		
	1	onPress	1
mc.onPress = function(mouseIdx:Number)			
{			
if (mouseIndex == 0)			
. . .			
else if (mouseIndex == 1)			
. . .			
. . .			
}			
onPress	onRelease		/
-1,	/	0	,
mc.onPress = function(mouseIdx:Number, keyboardOrMouse:Number)			
{			
if (keyboardOrMouse == 0)			
. . .			
else			

```

    . . .
}

    2   onRollOver/Out   onDragOver/Out
rollover/dragover                                     onRollOver/onRollOut
onDragOver/onDragOut                               onRollOver/onDragOver
    onRollOut/onDragOut                                     2
    onRollOver/onRollOut   onDragOver/onDragOut
        onRollOver/onDragOver           2   0
    onRollOver/onRollOut                       1
onRollOut/onDragOut           2       1       onRollOut/onDragOut
    2   0

    2           onRollOver/onRollOut   onDragOver/onDragOut
2

mc.onRollOver = function(mouseIdx:Number, nestingIdx:Number)
{
    //   nestingIdx == 0
    if (nestingIdx == 0)
        doOverAnimation();
    . . .
}
mc.onRollOut = function(mouseIdx:Number, nestingIdx:Number)
{
    //   nestingIdx == 0
    if (nestingIdx == 0)
        doOutAnimation();
    . . .
}

```

RollOver RollOut

onPress onRelease , :

- onPressAux
- onReleaseAux
- onReleaseOutsideAux
- onDragOver
- onDragOut

```

        ,
        index != 0 (0
    )
    (
    )
    ,
    .

```

```

mc.onPressAux = function(mouseIdx:Number, keyboardOrMouse:Number, buttonIdx:Number)
{
    if (buttonIdx == 1) // Right mouse button
        . . .
    . . .
}

```

2.3

ActionScript

Flash

Mouse.addListener

- onMouseMove = function() { }
- onMouseDown = function() { }
- onMouseUp = function() { }
- onMouseWheel = function(delta : Number, targetPath : String) { }

Flash OnMouseDown

onMouseUp

Scaleform

_global.gfxExtensions

- onMouseDown = function(button : Number, [targetPath : String],
[mouseIdx : Number], [x : Number], [y : Number],
[dblClick : Boolean]) { }
- onMouseUp = function(button : Number, [targetPath : String],
[mouseIdx : Number], [x : Number], [y : Number]) { }

Scaleform

RIGHT, MIDDLE

LEFT RIGHT MIDDLE
LEFT

1 2 3

16

Flash

- -
- - useHandCursor
- | -

Flash		Scaleform	C++ API	ActionScript
	C++	Gfx::StateBag::SetUserEventHandler		
		FxPlayer.cpp		
		ActionScript		
SDK		Mouse.flc/swf		
				‘ Cursor _M ’
		“ ” “ ” “ _ ”		
		_x _y		
			Scaleform	Mouse.setCursorType
		cursorType mouseIndex		
		(Mouse["ARROW"] Mouse["HAND"] and Mouse["IBEAM"])	cursorType	

```

_global.gfxExtensions = 1;
//
Mouse.hide();

var mouseListener:Object = new Object;
mouseListener.onMouseMove = function(mouseIdx, x,y)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    if (x == undefined)
    {
        x = _xmouse;
        y = _ymouse;
    }
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));

    cmc._x = x;
    cmc._y = y;
}
Mouse.addListener(mouseListener);

```

```

Mouse["setCursorType"] = function(cursorType, mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    switch(cursorType)
    {
        case Mouse["HAND"]:
            cmc.Cursor.gotoAndPlay("hand");
            break;
        case Mouse["ARROW"]:
            cmc.Cursor.gotoAndPlay("arrow");
            break;
        case Mouse["IBEAM"]:
            cmc.Cursor.gotoAndPlay("ibeam");
            break;
        default: return;
    }
}

//      Mouse.show      Mouse.hide
//
ASSetPropFlags(Mouse, "show,hide", 0, 7);

Mouse.show = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = true;
}
Mouse.hide = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = false;
}

```

2.5 ActionScript

Scaleform	ActionScript	ActionScript 1.0
	Mouse.setCursorType(...)	ActionScript 2.0
	ActionScript 2.0	

getButtonsState()

```
public function getButtonsState(mouseIndex : Number) : Number
```

Scaleform 2.2

1

```
mouseIndex : Number      -
```

getTopMostEntity()

```
public function getTopMostEntity([mouseIndex : Number]) : Object
public function getTopMostEntity(testAll : Boolean,
                                   [mouseIndex : Number]) : Object
public function getTopMostEntity(x : Number, y : Number) : Object
public function getTopMostEntity(x : Number, y : Number, testAll: Boolean) : Object
```

Scaleform 1.2.34, 2.2

onPress,

onMouseDown, onRollOver

```
MovieClip.hitTest      hitTest      x    y
getTopMostEntity      x    y      Mouse.getTopMostEntity(x,
y).hitTest(x, y, true) == true
```

```
mouseIndex : Number      -
testAll : Boolean      -      false      false
getTopMostEntity      ' true '
x : Number, y : Number -
```

:

```
var listenerObj = new Object;
```

```

listenerObj.onMouseMove = function()
{
    var target = Mouse["getTopMostEntity"]();
    trace("Mouse moved, target = "+target);
}

Mouse.addListener(listenerObj);
var target = Mouse["getTopMostEntity"](480, 10);
trace("The character at the (480, 10) is " + target);

```

getPosition

```
public function getPosition(mouseIndex : Number) : flash.geom.Point
```

Scaleform 2.2

_root

flash.geom.Point

mouseIndex : Number -

setCursorType()

```
public function setCursorType(cursorType : Number, [mouseIndex : Number]) : void
```

Scaleform 1.2.32

cursorType

“ ”

cursorType : Number -

Mouse.ARROW, Mouse.HAND, Mouse.IBEAM.

mouseIndex : Number -

3

hitTestDisable

hitTestDisable:Boolean [-]

Scaleform 2.1.51

true MovieClip.hitTest

false

TextField.hitTestDisable
MovieClip.hitTestDisable

topmostLevel

topmostLevel:Boolean [-]

Scaleform 2.1.50

true

false

- “topmostLevel” , :
Scaleform 3.0.71 , “topmostLevel”
“ ” (, “ ”);
 - Scaleform 3.0.72 , “ ” , A
B , A B , “topmostLevel”
“ ”
- : “topmostLevel”,swapDepth ActionScript

TextField.topmostLevel
MovieClip.topmostLevel

focusGroupMask

focusGroupMask : Number

Scaleform 3.3.84

(ALL)

```
,
setControllerFocusGroup
, " 1"(button1) 0 , " 2"(movieclip2) 0
1 ,
```

```
Selection.setControllerFocusGroup(0, 0);
Selection.setControllerFocusGroup(1, 1);
```

```
button1.focusGroupMask = 0x1; // bit 0 - focus group 0
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1
```

"focusGroupMask" (parent movieclip)

4 MovieClip

hitTest ()

```
public hitTest(x:Number, y:Number, [shapeFlag:Boolean],
               [ignoreInvisibleChildren:Boolean]):Boolean
```

Scaleform 2.1.51

```
3 hitTest
  "true" x y
_alpha 0
```

Flash hitTest
_visible false

```
ignoreInvisibleChildren:Boolean - true
```

hitTestDisable

```
hitTestDisable:Boolean [ - ]
```

Scaleform 1.2.32

```
true MovieClip.hitTest
```

```
false
```

```
TextField.hitTestDisable
Button.hitTestDisable
```

noAdvance

```
noAdvance : Boolean
```

Scaleform 2.1.52

ActionScript

Scaleform Flash “ true ”

```
_global.noInvisibleAdvance
```

topmostLevel

```
topmostLevel:Boolean  [ - ]
```

Scaleform 2.1.50

true

false

```
“topmostLevel”      ,      :
```

- Scaleform 3.0.71 , “topmostLevel”
“ ” (, “ ”);
- Scaleform 3.0.72 , “ ” , A
B , A B , “topmostLevel”
“ ”

```
: "topmostLevel",swapDepth ActionScript
```

TextField.topmostLevel

Button.topmostLevel

rendererString

```
rendererString:String  [ - ]
```

Scaleform 2.2.55

MovieClip

ActionScript

```
myMovieInstance.rendererString = "SHADER_Blur"
```

rendererFloat

rendererFloat:Number [-]

Scaleform 2.2.55

MovieClip

ActionScript

```
myMovieInstance.rendererFloat = 4; // C++
```

disableBatching

disableBatching:Boolean

Scaleform : 4.0.17

focusGroupMask

focusGroupMask : Number

Scaleform 3.3.84

(ALL)

,
setControllerFocusGroup

```

    , "1"(button1) 0 , "2"(movieclip2) 0
1
,
Selection.setControllerFocusGroup(0, 0);
Selection.setControllerFocusGroup(1, 1);

button1.focusGroupMask = 0x1; // bit 0 - focus group 0
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1

"focusGroupMask" (parent movieclip)

```

5 NetStream

Scaleform NetStream

onMetaData

```
onMetaData = function(info:Object) { }
```

Scaleform 3.0.63

Scaleform

onMetaData

1. audioTracks -

audioTracks

channelsNumber - (1- , 2- , 6-5.1)

totalSamples -

trackIndex -

NetStream.audioTrack ()

sampleRate -

2. subtitleTracksNumber -

audioTrack: [-]

Scaleform 3.0.63

/

```
var ns:NetStream = new NetStream(nc);  
var audioTracks;
```

```
ns.onMetaData = function(info:Object)
```

```

{
    audioTracks = info.audioTracks;
}

if (audioTracks != undefined && audioTracks.length > 0)
    ns.audioTrack = audioTracks[0].trackIndex;

```

subtitleTrack: Number [-]

Scaleform 3.0.63

0

onSubtitle

```
onSubtitle = function(msg:String) {}
```

Scaleform 3.0.63

```

var ns:NetStream = new NetStream(nc);
var subtitlesNumber = 0;

ns.onMetaData = function(info:Object)
{
    subtitlesNumber = info.subtitleTracksNumber;
}

ns.onSubtitle = function(msg:String)
{
    sbTextField.text = msg;
}
if (subtitlesNumber > 0)
    ns.subtitleTrack = 1;

```

setNumberOfFramePools

```
public function setNumberOfFramePools(pools : Number) : Void
```

Scaleform 3.0.68

CPU “ ”
1
NetStream.play()

setReloadThresholdTime

```
public function setReloadThresholdTime(reloadTime : Number) : Void
```

Scaleform 3.0.68

Scaleform 0.8
Netstream 1
NetStream.setBufferTime() 2 1
“ ”
NetStream.play()

6

Scaleform	ActionScript 1.0
Selection.captureFocus()	ActionScript 2.0
ActionScript 2.0	Selection["captureFocus"]()

alwaysEnableArrowKeys

alwaysEnableArrowKeys : Boolean

Scaleform 1.2.34

“_focusrect ”	false
“_focusrect = false ”	Flash
alwaysEnableArrowKeys	“true ”

```
Selection["alwaysEnableArrowKeys"] = true;
```

alwaysEnableKeyboardPress

alwaysEnableKeyboardPress : Boolean

Scaleform 3.0.63

	onPress / onRelease	focusrect
false		focusrect = false
Flash		alwaysEnableKeyboardPress
true		

```
Selection["alwaysEnableKeyboardPress"] = true;
```

captureFocus()

```
public function captureFocus([doCapture:Boolean, controllerIdx:Number]) : void
```

Scaleform 1.2.34

```
captureFocus doCapture true
              Tab
captureFocus doCapture false
```

```
doCapture:Boolean –
                    captureFocus true
controllerIdx:Number – , / , 0
```

```
Selection["captureFocus"](); // “true”
Selection["captureFocus"](false);
```

disableFocusAutoRelease

```
disableFocusAutoRelease : Boolean
```

Scaleform 1.2.34

Flash true

```
Selection["disableFocusAutoRelease"] = true;
```


disableFocusKeys

disableFocusKeys : Boolean

Scaleform 2.2.58

TAB, Shift-TAB

```
Selection["disableFocusKeys"] = true;
```

```
moveFocus()
```

disableFocusRolloverEvent

disableFocusRolloverEvent : Boolean

Scaleform 2.0.37

rollover/out true

```
Selection["disableFocusRolloverEvent "] = true;
```

modalClip

modalClip : MovieClip

Scaleform 2.2.58

“ modal ”

TAB,Shift-TAB
“ tabable ”

```
Selection["modalClip"] = _root.mc;
```

```
...
Selection["modalClip"] = undefined;
```

```
disableFocusKeys
moveFocus()
```

moveFocus()

```
public function moveFocus(keyToSimulate : String [, startFromMovie:Object,
    includeFocusEnabledChars : Boolean = false,
    controllerIdx :Number]) : Object
```

Scaleform : 2.2.58

TAB, Shift-TAB

disableFocusKeys modalClip

```
keyToSimulate:String - "up", "down", "left", "right", "tab", "shifftab"
startFromMovie:Object - moveFocus
                        null
```

```
includeFocusEnabledChars:Boolean - moveFocus
                                focusEnabled
                                tabEnabled / tabIndex
                                false
```

```
                                tabEnabled / tabIndex
controllerIdx:Number - , / , 0
```

```
Selection["moveFocus"]("up");
Selection["moveFocus"]("tab", _root.mc);
Selection["moveFocus"]("tab", _root.mc, true);
Selection["moveFocus"]("tab", null, true);
```

```

disableFocusKeys
modalClip

```

findFocus()

```

public function findFocus(keyToSimulate : String [, parentMovie:Object, loop :
    Boolean, startFromMovie:Object, includeFocusEnabledChars : Boolean,
    controllerIndex : Number]) : Object

```

Scaleform 3.3.84

:TAB Shift-TAB

disableFocusKeys setModalClip/getModalClip

```

keyToSimulate:String- " " " " " " " " " " " "
"shifftab"
parentMovie - " "(Null)
loop - " , findFocus "null"( "false"( ) ),
( "true"( ) )
startFromMovie:Object - , findFocus
"null" " undefined"( ),

includeFocusEnabledChars:Boolean - , moveFocus
focusEnabled tabEnabled/tabIndex
false( ),
tabEnabled/tabIndex
controllerIndex -

```

```

var a = Selection["findFocus"]("up");

```

, null ---

```

disableFocusKeys
setModalClip
getModalClip

```

setModalClip()

```
public function setModalClip(modalClip : Object, controllerIndex : Number)
```

Scaleform 3.3.84

```

    "tabable"
    " " TAB Shift-TAB

```

```

modalClip -
controllerIndex -

```

getModalClip()

```
public function getModalClip(controllerIndex : Number) : Object
```

Scaleform 3.3.84

```
controllerIndex -
```

```
, ---
```

setControllerFocusGroup()

```
public function setControllerFocusGroup (controllerIndex : Number, focusGroupIdx :
Number) : Boolean
```

Scaleform 3.3.84

```

controllerIndex , 0
,
( ), setControllerFocusGroup (1,1) 1
setControllerFocusGroup (1,0) 0 1

```

```

controllerIndex -
focusGroupIdx -

```

```

Selection["setControllerFocusGroup"](0,0);
Selection["setControllerFocusGroup"](1,1);
Selection["setControllerFocusGroup"](2,1);

```

true ---

getControllerFocusGroup()

public function getControllerFocusGroup (controllerIndex : Number) : Number

Scaleform 3.3.84

controllerIndex -

getFocusArray()

public function getFocusArray(mc : Object) : Array

Scaleform 3.3.84

, / /

mc -

()

getFocusBitmask()

public function getFocusBitmask(mc : Object) : Number

Scaleform 3.3.84

, / /

mc -

getControllerMaskByFocusGroup()

```
public function getControllerMaskByFocusGroup (focusGroupIdx :Number) :Number
```

Scaleform 3.3.84

```
setControllerFocusGroup ,
```

```
focusGroupIdx -
```

numFocusGroups

```
numFocusGroups : Number
```

Scaleform 3.3.84

```
 , setControllerFocusGroup 0 3 ,  
numFocusGroups 2
```

7 TextField

Scaleform TextField Flash 8 TextField
 Flash 9 TextField IME

7.1

autoFit

autoFit:Boolean [-]

Scaleform 2.0.39

 on/off

 false

appendText ()

public function appendText(newText:String):void

Scaleform 2.0.37

 newText +=

my_txt.text += appendingText

 : ,

 newText:String –

 appendHtml()

appendHtml ()

public function appendHtml(newHtml:String):void

Scaleform 2.0.37

newHtml HTML
txt.htmlText += moreHtml htmlText += HTML HTML
HTML
newHtml HTML += HTML
:
,

newHtml:String - HTML

appendText()

caretIndex

caretIndex:Number [-]

Scaleform 2.0.37

0

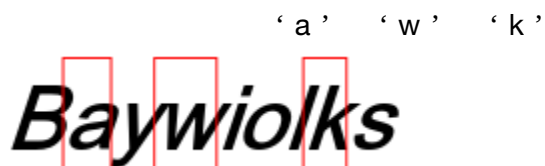
Selection.getCaretIndex() caretIndex
Selection.getCaretIndex() -1
0

Selection.getCaretIndex()

getCharBoundaries ()

public function getCharBoundaries(charIndex:Number): flash.geom.Rectangle

Scaleform 2.0.37



charIndex:Number – 0 1

flash.geom.Rectangle – x y
0 0

getExactCharBoundaries()

getExactCharBoundaries ()

public function getExactCharBoundaries(charIndex:Number): flash.geom.Rectangle

Scaleform 2.0.37

getCharBoundaries()
‘ a ’ ‘ w ’ ‘ k ’



charIndex: Number - 0 1

flash.geom.Rectangle - x y
0 0

getCharBoundaries()

getCharIndexAtPoint ()

public function getCharIndexAtPoint(x:Number, y:Number): Number

Scaleform 2.0.37

x y

x: Number - x
y: Number - y

Number - 0 1
-1

getFirstCharInParagraph ()

public function getFirstCharInParagraph(charIndex: Number): Number

Scaleform 2.0.37

charIndex

charIndex: Number - 0 1

Number -

getLineIndexAtPoint ()

```
public function getLineIndexAtPoint(x:Number, y:Number): Number
```

Scaleform 2.0.37

x y

x:Number - x

y:Number - y

Number - 0 1
-1

getLineLength()

```
public function getLineLength(lineIndex:Number): Number
```

Scaleform 2.0.37

lineIndex:Number - 0 1

Number -

getLineMetrics()

```
public function getLineMetrics(lineIndex:Number): Object
```

Scaleform 2.0.43

ascent : Number

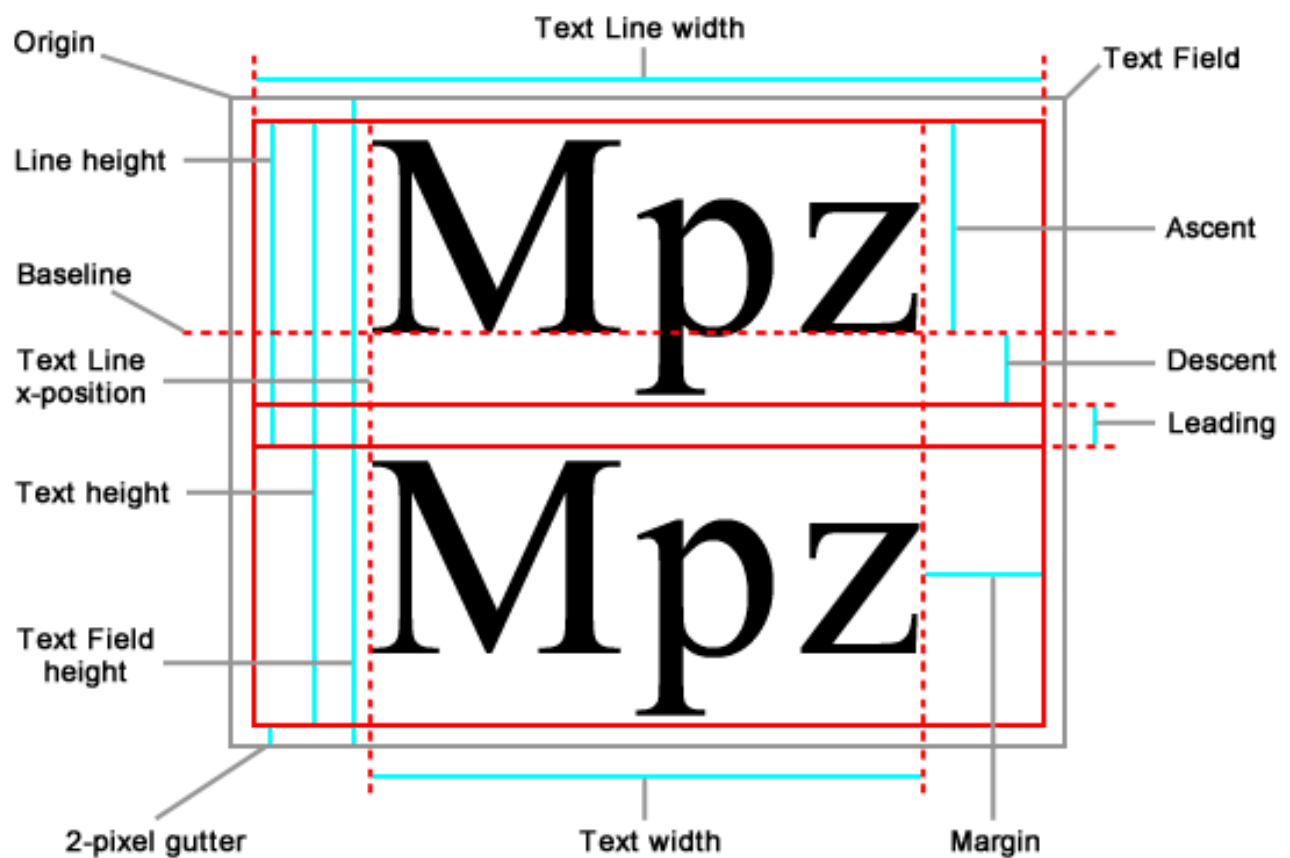
descent : Number

height : Number

leading : Number

width : Number

x : Number



lineIndex: Number - 0 1

Number -

getLineOffset ()

public function getLineOffset(lineIndex: Number): Number

Scaleform 2.0.37

lineIndex

lineIndex: Number - 0 1

Number -

getLineText ()

public function getLineText(lineIndex: Number): String

Scaleform 2.0.43

lineIndex

lineIndex: Number - 0 1

String -

hitTestDisable

hitTestDisable: Boolean [-]

Scaleform 1.2.32

true MovieClip.hitTest

false

noTranslate

noTranslate:Boolean [-]

Scaleform 1.2.34

true GfxTranslator::Translate

false

numLines

numLines:Number [-]

Scaleform 2.0.43

wordWrap true

topmostLevel

topmostLevel:Boolean [-]

Scaleform 2.1.50

true

false

- "topmostLevel" , :
Scaleform 3.0.71 , "topmostLevel"
" " (, " ");
- Scaleform 3.0.72 , , " " , A
B , A B , "topmostLevel"
" "

: "topmostLevel",swapDepth ActionScript

MovieClip.topmostLevel
Button.topmostLevel

focusGroupMask

focusGroupMask : Number

Scaleform 3.3.84

(ALL)

,
setControllerFocusGroup
1, " 1"(button1) 0, " 2"(movieclip2) 0

Selection.setControllerFocusGroup(0, 0);
Selection.setControllerFocusGroup(1, 1);

button1.focusGroupMask = 0x1; // bit 0 - focus group 0
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1

"focusGroupMask" (parent movieclip)

7.2

alwaysShowSelection

alwaysShowSelection:Boolean [-]

Scaleform 2.1.45

Scaleform true
Scaleform
” “

false

selectionBkgColor
selectionTextColor
inactiveSelectionBkgColor
inactiveSelectionTextColor

copyToClipboard ()

```
public function copyToClipboard([richTextClipboard:Boolean], [startIndex:Number],  
[endIndex:Number]):void
```

Scaleform 2.1.45

richTextClipboard:Boolean –	true	
	useRichTextClipboard	
startIndex:Number –		selectionBeginIndex
endIndex:Number –		selectionEndIndex
useRichTextClipboard		
selectionBeginIndex		
selectionEndIndex		
cutToClipboard()		
pasteFromClipboard()		

cutToClipboard ()

```
public function cutToClipboard([richTextClipboard:Boolean], [startIndex:Number],  
[endIndex:Number]):void
```

Scaleform 2.1.45

richClipboard:Boolean –	true	
	useRichTextClipboard	
startIndex:Number –		selectionBeginIndex
endIndex:Number –		selectionEndIndex
useRichTextClipboard		
selectionBeginIndex		
selectionEndIndex		
copyToClipboard()		
pasteFromClipboard()		

inactiveSelectionBkgColor

inactiveSelectionBkgColor:Number [-]

Scaleform 2.1.45

		alwaysShowSelection	true
0xAARRGGBB	AA	alpha	[0,255] RR -
[0,255] BB -		[0,255] GG -	[0,255]
	alpha	0	alpha 0
	Flash		
0xFFFF0000	alpha	0xFF (255)	“backgroundColor”
0xFF0000			

alwaysShowSelection
inactiveSelectionTextColor
selectionBkgColor
selectionTextColor

inactiveSelectionTextColor

inactiveSelectionTextColor: Number [-]

Scaleform 2.1.45

		alwaysShowSelection	true	
0xRRGGBB	AA	alpha	[0,255]	RR - [0,255]
BB -		[0,255]	GG -	[0,255]
	alpha	0		alpha 0
		Flash		
0xFFFF0000	alpha	0xFF (255)	"textColor"	0xFF0000

alwaysShowSelection
inactiveSelectionBkgColor
selectionBkgColor
selectionTextColor

noAutoSelection

noAutoSelection: Boolean [-]

Scaleform 2.1.45

true

false

pasteFromClipboard ()

public function pasteFromClipboard([richClipboard: Boolean], [startIndex: Number], [endIndex: Number]): void

Scaleform 2.1.45

richClipboard:Boolean –	true	
	useRichTextClipboard	
startIndex:Number –		selectionBeginIndex
endIndex:Number –		selectionEndIndex

```

useRichTextClipboard
selectionBeginIndex
selectionEndIndex
copyToClipboard()
cutToClipboard()

```

selectionBeginIndex

selectionBeginIndex:Number [-]

Scaleform 2.1.45

caretIndex

Selection.getBeginIndex()	
Selection.getBeginIndex()	Selection.getBeginIndex()
-1	

```

caretIndex
selectionEndIndex
Selection.getBeginIndex()

```

selectionEndIndex

selectionEndIndex:Number [-]

Scaleform 2.1.45

caretIndex

Selection.getEndIndex()	Selection.getEndIndex()
Selection.getEndIndex()	-1

caretIndex
selectionBeginIndex
Selection.getEndIndex()

selectionBkgColor

selectionBkgColor: Number [-]

Scaleform 2.1.45

		0xRRGGBB	AA	alpha	
[0,255]	RR -	[0,255]	BB -	[0,255]	GG -
	[0,255]				
	alpha			alpha	0
		Flash			
0xFFFF0000	alpha	0xFF (255)	BkgColor		0xFF0000

inactiveSelectionTextColor
inactiveSelectionBkgColor
selectionTextColor

selectionTextColor

selectionTextColor: Number [-]

Scaleform 2.1.45

		0xRRGGBB	AA	alpha	
[0,255]	RR -	[0,255]	BB -	[0,255]	GG -
	[0,255]				
	alpha			alpha	0
		Flash			
0xFFFF0000	alpha	0xFF (255)	textColor		0xFF0000

inactiveSelectionBkgColor
inactiveSelectionBkgColor
selectionBkgColor

useRichTextClipboard

useRichTextClipboard:Boolean [-]

Scaleform 2.1.45

true Scaleform

useRichTextClipboard true

false

7.3

Scaleform

fontScaleFactor

fontScaleFactor:Number [-]

Scaleform 2.0.43

fontScaleFactor

1.0

textAutoSize

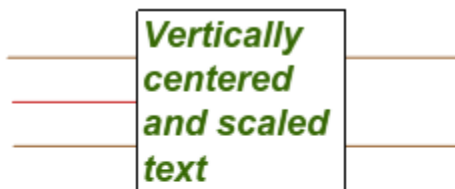
textAutoSize:String [-]

Scaleform 2.0.43

verticalAutoSize:String [-]

Scaleform 2.0.43

	“ none ”	“ top ”	“ bottom ”	“ center ”
verticalAutoSize	“ none ”			
verticalAutoSize	“ top ”	TextField.autoSize	wordWrap	
verticalAutoSize	“ center ”			
verticalAutoSize	“ bottom ”			



“ none ”

verticalAlign
TextField.autoSize

7.4 HTML

Scaleform HTML

 ALPHA

ALPHA="#xx"

alpha	00 FF	0-255	0	255	COLOR
ALPHA	HTML				

Scaleform 2.0/2.1		Scaleform	Web
(	"http://", "file://"		

HTML		" myImage "
------	-------	-------------

```
t.htmlText = "<p align='right'>abra<img src='myImage' width='20' height='30'
align='baseline' vspace='-10'>bed232</p>";
```

IMG

```
src - ;
(JPEG GIF PNG SWF ) "img://",
; , Gfx::ImageCreator::
```

LoadImage。

width -

height -

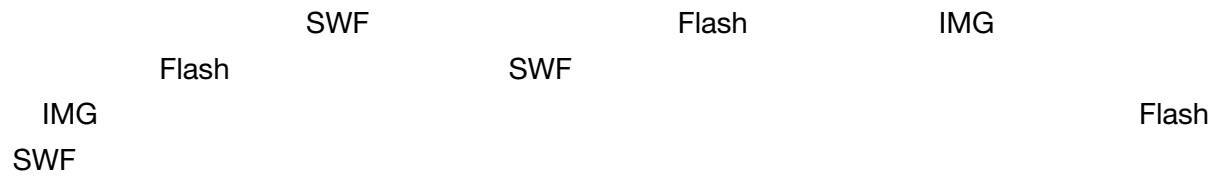
align - " baseline "

vspace - " baseline "

id -	Flash Player	SWF
------	--------------	-----

align - " left " " right "

hspace - " left " " right "



7.5

blur knockout

blurX, blurY

blurX:Number [-]

blurY:Number [-]

Scaleform 2.0.41

blur 0 15.9()

0

blurStrength

blurStrength

blurStrength:Number [-]

Scaleform 2.0.41

blur 0 15.9

0

blurX
blurY

shadowAlpha

shadowAlpha: Number [-]

Scaleform 2.0.41

alpha 0.00 1.00 0.25 25%

shadowColor

shadowAngle

shadowAngle: Number [-]

Scaleform 2.0.41

DropShadowFilter 0 360° ()

45

DropShadowFilter

shadowBlurX, shadowBlurY

shadowBlurX: Number [-]

shadowBlurY: Number [-]

Scaleform 2.0.41

blur 0 15.9()

0

shadowStrength

shadowColor

shadowColor: Number [-]

Scaleform 1.1.31

shadowStyle 0xRRGGBB RR -
[0..255] BB - [0..255] GG - [0..255]

shadowStyle

shadowDistance

shadowDistance: Number [-]

Scaleform 2.0.41

DropShadowFilter

DropShadowFilter

shadowHideObject

shadowHideObject: Boolean [-]

Scaleform 2.0.41

true false

shadowKnockOut

shadowKnockOut: Boolean [-]

Scaleform 2.0.41

true false

shadowQuality

shadowQuality:Number [-]

Scaleform 2.0.41

Flash

1

2

1

2

shadowStrength

shadowStrength:Number [-]

Scaleform 2.0.41

0-15.9

0

shadowBlurX

shadowBlurY

shadowStyle

shadowStyle:String [-]

Scaleform 1.1.31

shadowColor

shadowStyle

```
field.shadowStyle = "s{0,-1.5}{-1.4,1.2}{1.4,1.2}t{0,0}";  
field.shadowColor = 0xff0000;
```

“s”

shadowColor

“t”

TextField.textColor

“t”

“t 0 0

shadowColor
TextField.textColor

7.6

setImageSubstitutions()

```
public setImageSubstitutions(substInfoArr:Array) : void  
public setImageSubstitutions(substInfo:Object) : void  
public setImageSubstitutions(null) : void
```

Scaleform 2.0.38

SWF

SWF

- 1.
2. Windows Macintosh
3. ActionScript myImage
4. OK

BitmapData

ActionScript

```
import flash.display.BitmapData;  
var imageBmp:BitmapData = BitmapData.loadBitmap("myImage");
```

```
( import flash.display.BitmapData;  
flash.display.BitmapData) " "
```

ID

subString:String

15

image : BitmapData

width : Number

height : Number

baseLineY : Number

Y

w/o

id : String

ID “updateImageSubstitution ”

“substInfoArr ”

“substInfo ”

setImageSubstitutions(substInfo:Object)

setImageSubstitutions(substInfoArr:Array)

setImageSubstitutions

setImageSubstitutions(null)

setImageSubstitutions

ActionScript

ActionScript

substInfoArr:Array –

substInfo:Object –

null –

updateImageSubstitution()

```

var b1 = BitmapData.loadBitmap("smile1");
var b2 = BitmapData.loadBitmap("smile2");
var b3 = BitmapData.loadBitmap("smile3");
var a = new Array;
a[0] = { subString:"=", image:b1, baseLineY:35, width:20, height:20, id:"sm=" } };
a[1] = { subString":-)", image:b2, baseLineY:20, id:"sm:-)" } };
a[2] = { subString":\\", image:b3, baseLineY:35, height:100 } };
a[3] = { subString":-\\", image:b1 } };
t.setImageSubstitutions(a);

```

“ = ”

“ smile1 ”

updateImageSubstitution()

```
public updateImageSubstitution(id:String, image:BitmapData) : void
```

Scaleform 2.0.38

setImageSubstitutions

id:String –	ID	setImageSubstitutions
	ID	
image:BitmapData –		null

setImageSubstitutions()

```
t.updateImageSubstitution("sm=", b3);
```

onEnterFrame setInterval

updateImageSubstitution

```

var phase = 0;
var b1a = BitmapData.loadBitmap("smile1a");
var b2a = BitmapData.loadBitmap("smile2a");

onEnterFrame = function()
{

```

```
if (phase % 10 == 0)
{
    if (phase % 20 == 0)
    {
        _root.t.updateImageSubstitution("sm=", b1);
        _root.t.updateImageSubstitution("sm:-)", b2);
    }
    else
    {
        _root.t.updateImageSubstitution("sm=", b1a);
        _root.t.updateImageSubstitution("sm:-)", b2a);
    }
}
++phase;}
```


7.7 IME

IME

disableIME

disableIME:Boolean [-]

Scaleform 2.1.50

true IME

false

getIMECompositionStringStyle()

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform 2.1.50

IME

setIMECompositionStringStyle

categoryName

categoryName:String – IME

setIMECompositionStringStyle

-

setIMECompositionStringStyle

setIMECompositionStringStyle

disableIME

setIMECompositionStringStyle()

```
public function setIMECompositionStringStyle(categoryName:String,
                                             styleDescriptor:Object): Number
```

Scaleform 2.1.50

IME categoryName

```
"compositionSegment" –
"clauseSegment" –
"convertedSegment" –
"phraseLengthAdj" –
"lowConfSegment" –
```

styleDescriptor

textColor : Number

backgroundColor : Number

underlineColor : Number

underlineStyle : String

```
“single”
“thick”
“dotted”
“ditheredSingle”
“ditheredThick”
```

```
[0..255] GG- 0xRRGGBB RR- [0..255] BB-
[0..255]
```

```
categoryName:String – IME
styleDescriptor:Object –
```

getIMECompositionStringStyle
disableIME

8 **TextFormat**

TextFormat

alpha

alpha:Number [-]

Scaleform 2.0.41

	alpha	0-100%	Flash TextFormat.color
alpha			

TextFormat

HTML

9

Stage

Scaleform

“ visibleRect ”

“ safeRect ”

“ originalRect ”

“ onResize ”

ActionScript 1.0

Stage.visibleRect

ActionScript 2.0

ActionScript 2.0

```
Stage["visibleRect"]
```

visibleRect

```
visibleRect:flash.geom.Rectangle [ - ]
```

Scaleform 2.2.56

Viewport

topLeft

safeRect

originalRect

safeRect

```
safeRect:flash.geom.Rectangle [ - ]
```

Scaleform 2.2.56

Scaleform::Movie::SetSafeRect

“ visibleRect ”

visibleRect

originalRect

originalRect

`originalRect:flash.geom.Rectangle [ - ]`

Scaleform 2.2.56

SWF
600px×400px

SWF

Flash Studio
{ 0, 0, {600, 400} }

“ Document properties ”

`visibleRect`
`safeRect`

onResize

`onResize = function([visibleRect:flash.geom.Rectangle]) {}`

Scaleform 2.2.56

`onResize`

`Stage.visibleRect`

`visibleRect`

translateToScreen()

`public function translateToScreen(pt:Object): Point`

Scaleform 3.3.84

Stage

`pt:Object –`
`flash.geom.Point,`

`Object`

`x y`

`Object`

–

Flash	Array.sort	Array.sortOn	Unicode
			Unicode ‘ z’
			Flash [‘á’, ‘z’, ‘a’]
[‘a’, ‘z’, ‘á’]			
	Scaleform		
		Array.sort/sortOn	
Array.LOCALE:			
var arr = [‘á’, ‘z’, ‘a’]; var newArr1 = arr.sort(); // [‘a’, ‘z’, ‘á’] var newArrLoc = arr.sort(Array.LOCALE); // [‘a’, ‘á’, ‘z’] var newArrRev = arr.sort(Array.LOCALE Array.DESENDING); // [‘z’, ‘á’, ‘a’]			
var arr = [‘Á’, ‘z’, ‘a’]; var a = arr.sort(Array.LOCALE Array.CASEINSENSITIVE); // [‘a’, ‘Á’, ‘z’]			
			Array.LOCALE
	Unicode		
Scaleform	3.0.65		
String.localeCompare			

11 String

localeCompare()

0

“ ”

Unicode

```
public localeCompare(other:String [, caseInsensitive:Boolean]) : Number
```

Scaleform 3.0.65

other:String –
caseInsensitive:Boolean –

Array.sort

12

Flash

NetStream

Scaleform

NetStream

clipRect

clipRect: flash.geom.Rectangle [-]

Scaleform 3.0.70

```
video.clipRect = new flash.geom.Rectangle(10, 20, 100, 100);
```

13 3Di Extensions

Scaleform 3Di 是使用一組內置到 Scaleform 中的基本 AS2 擴展實現的。為 3Di 增加了下列 ActionScript 擴展：

Scaleform 3.2.82

_z

Z (), 0

_zscale

將 Z 向的物件伸縮設置一個百分比,預設值為 100。

_xrotation

X(), 0

_yrotation

Y(), 0

_matrix3d

使用一組 16 個浮點數(4 x 4 矩陣)設置物件的完整 3D 轉換。如果此值設置為 NULL,就會刪除所有 3D 轉換,而且物件將會變為 2D。

perspfov

在物件上設置透視視野 (Field Of View) 角度,有效角度必須 > 0 度和 < 180 度。180="">></ 180 度。>如果不設置,物件就繼承根 FOV 角,該角預設為 55 度。

使用這些新擴展需要在 ActionScript 中把全域變數 gfxExtensions 設置為 True(真)。

14

noInvisibleAdvance

noInvisibleAdvance : Boolean

Scaleform 2.1.52

true
Flash

“true”

Scaleform Flash

SWF
ActionScript

MovieClip.noAdvance

imecommand()

imecommand(command:String, parameters:String) : Void

Scaleform 2.1.50

fscommand

Scaleform IME

Scaleform

getIMECandidateListStyle()

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform 2.1.50

IME

setIMECandidateListStyle

Object –

setIMECandidateListStyle
TextField.disableIME

setIMECandidateListStyle()

public function setIMECandidateListStyle(styleDescriptor:Object)

Scaleform 2.1.50

styleDescriptor

styleObject.textColor	= 0x5EADFF;
styleObject.selectedTextColor	= 0xFFFFFFFF;
styleObject.fontSize	= 20;
styleObject.backgroundColor	= 0x001430;
styleObject.selectedTextBackgroundColor	= 0x2C5A99;
styleObject.indexBackgroundColor	= 0x12325A;
styleObject.selectedIndexBackgroundColor	= 0x2C5A99;
styleObject.readingWindowTextColor	= 0xFFFFFFFF;
styleObject.readingWindowBackgroundColor	= 0x001430;
styleObject.readingWindowFontSize	= 20;

textColor : Number

selectedTextColor : Number

fontSize : Number

backgroundColor : Number

seletedTextBackgroundColor : Number

indexBackgroundColor : Number

selectedIndexBackgroundColor : Number

readingWindowTextColor : Number

readingWindowBackgroundColor : Number

readingWindowFontSize : Number

0xRRGGBB RR - [0,255] BB -
[0,255] GG - [0,255]

getIMECandidateListStyle
TextField.disableIME

15

Key

```
Key.getCode(controllerIdx:Number)
Key.getAscii(controllerIdx:Number)
Key.isDown(controllerIdx:Number)
Key.isToggled(controllerIdx:Number)
```

```
Key / controllerIdx 0
```

```
Key.onKeyDown(controllerIdx:Number)
```

```
Key onKeyDown /
```

Selection

```
Selection.setFocus(ch:Object, controllerIdx:Number)
Selection.setFocus(controllerIdx:Number)
Selection.getBeginIndex(controllerIdx:Number)
Selection.getEndIndex(controllerIdx:Number)
Selection.setSelection(start:Number, end:Number, controllerIdx:Number)
Selection.getCaretIndex(controllerIdx:Number)
```

```
Selection / , controllerIdx 0
```

```
Selection.onSetFocus(old:Object, new:Object, controllerIdx:Number)
```

```
Selection onSetFocus /
```

MovieClip/Button/TextField

```
MovieClip.onSetFocus(old:Object, controllerIdx:Number)
MovieClip.onKillFocus(new:Object, controllerIdx:Number)
Button.onSetFocus(old:Object, controllerIdx:Number)
Button.onKillFocus(new:Object, controllerIdx:Number)
TextField.onSetFocus(old:Object, controllerIdx:Number)
TextField.onKillFocus(new:Object, controllerIdx:Number)
TextField.onChange(controllerIdx:Number)
```

```
MovieClip/Button/TextField /
```

System.capabilities

```
System.capabilities.numControllers -
```